FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 183**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

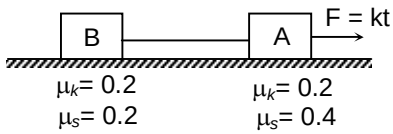
Physics

PART – I

SECTION – A
(One or More than one correct type)

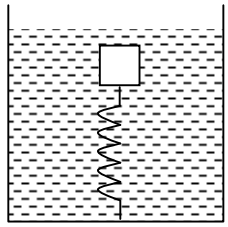
This section contains **7** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Two block *A* and *B* each of mass $\frac{1}{2}$ kg is connected by a massless inextensible string and kept on horizontal surface. Coefficient of friction between block and surface shown in figure. A force $F = kt$ (where k = 1 N/s and t is time in second) applied on *A.* Then (g = 10 m/s^2)



(A) work done by friction force on block *B* is zero in time interval t = 0 to t = 3s.
(B) work done by friction force on block *A* is zero in time interval t = 0 to t = 3s.
(C) work done by tension on *B* is also zero in time interval t = 0 to t = 3s.
(D) speed of blocks at t = 10s is 27.5 m/s.

2. A block of mass '*m*' is attached by means of a spring to the bottom of a tank of water as shown in figure. At equilibrium, the spring is under compression. If the tank is now allowed to fall freely, then choose the correct alternative(s)



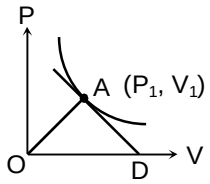
(A) the spring comes to its relaxed position
(B) the spring compresses more than its equilibrium compression
(C) the boyant force becomes zero
(D) there will be some elongation is the spring

Space for Rough work

3. A particle of mass *m* moves in a straight line. If *v* is the velocity at a distance *x* from a fixed point on the line and $v^2 = a - bx^2$, where *a* and *b* are constant, then
(A) the motion continues along the positive *x*-direction only
(B) the motion is simple harmonic\
(C) the particle oscillates with a frequency equal to $\frac{\sqrt{b}}{2\pi}$
(D) the total energy of the particle is *ma*

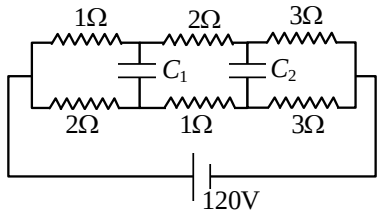
4. A traveling wave pulse is given by $y = \frac{6}{2 + (x + 3t)^2}$ where symbols have their usual meanings, *x, y* are in metre and t is in second. Then
(A) the pulse is traveling along +ve *x*-axis with velocity 3 m/s
(B) the pulse is traveling along –ve *x*-axis with velocity 3 m/s
(C) the amplitude of the wave pulse is 3 m
(D) the pulse is a symmetric pulse

5. *n* moles of an ideal gas undergo an isothermal process at temperature *T*. P-V graph of the process is as shown in the figure. A point *A* (V_1, P_1) is located on the P-V curve. Tangent at point *A*, cuts the *V*-axis at point *D*. *AO* is the line joining the point *A* to the origin *O* of PV diagram. Then,



(A) coordinates of points *D* is $\left(\frac{3V_1}{2}, 0\right)$
(B) coordinates of points *D* is $(2V_1, 0)$
(C) area of the triangle *AOD* is *nRT*
(D) area of the triangle *AOD* is $\frac{3}{4}nRT$

6. In the circuit shown in figure $C_1 = C_2 = 2\mu F$. Then charge stored in
(A) capacitor C_1 is zero
(B) capacitor C_2 is zero
(C) both capacitors is zero
(D) capacitor C_1 is 40 μC



Space for Rough work

7. A converging lens of focal length f_1 is placed in front of and coaxially with the convex mirror of focal length f_2. Their separation is d. A parallel beam of light incident on the lens returns as a parallel beam from the arrangement
(A) The beam diameters of the incident and reflected beams must be the same
(B) $d = |f_1| - 2|f_2|$
(C) $d = |f_2| - |f_1|$
(D) If the entire arrangement immersed in water, the conditions will remain unaltered

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

A container of cross section area A_1 and height L is filled with a liquid of density d_1 up to height 4L/5. A cylinder of length L, cross section area A_2 and density d_2 is submerged into the container gently so that it is floating. ℓ is height of liquid in container after cylinder is submerged.

Column 1	Column 2	Column 3
(I) $d_1 = 2d$ $d_1 = 2d$	(i) $\ell = 0.88L$	(P) $A_1 = 12A$ $A_2 = 2A$
(II) $d_1 = d$ $d_2 = d$	(ii) $\ell = L$	(Q) $A_1 = 10A$ $A_2 = A$
(III) $d_1 = d$ $d_2 = 5d$	(iii) $\ell = 0.85L$	(R) $A_1 = 5A$ $A_2 = A$
(IV) $d_1 = 3d$ $d_2 = 9d$	(iv) $\ell = 0.82L$	(S) $A_1 = 18A$ $A_2 = 3A$

8. In which case would the cylinder be floating completely submerged.
(A) (I) (ii) (S) (B) (I) (iii) (R)
(C) (II) (ii) (R) (D) (II) (iii) (Q)

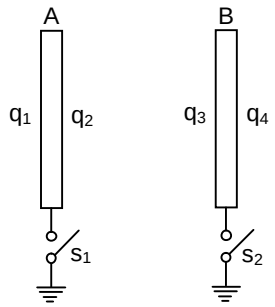
Space for Rough work

9. In which case the change in pressure at the bottom of the container is minimum?
(A) (III)(ii) (S) (B) (I) (iv) (Q)
(C) (III) (iv) (Q) (D) (II) (iii) (Q)

10. In which case the liquid will exert maximum force on the cylinder?
(A) (IV)(ii) (P) (B) (I) (iv) (S)
(C) (IV) (iii) (S) (D) (II) (iii) (Q)

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

In the figure given below, plate A has 100μC charge, while plate B has 60 μC charge. Switches S_1 and S_2 are connected to the ground. q_1, q_2, q_3 and q_4 are charges on faces of plates as shown in figure



Column 1	Column 2	Column 3
(I) q_1 =80 μC	(i) q_2= 100 μC	(P) q_3= -20 μC
(II) q_1 = 60 μC	(ii) q_2= 0 μC	(Q) q_3= -100 μC
(III) q_1 = 0 μC	(iii) q_2= 20 μC	(R) q_3= 60 μC
(IV) q_1 = -60 μC	(iv) q_2= 60 μC	(S) q_3= 0 μC

11. Which of the following options correctly represents the case when switch S_1 is closed and S_2 is open?
(A) (IV)(ii) (P) (B) (III) (i) (R)
(C) (III) (iv) (S) (D) (II) (iii) (Q)

Space for Rough work

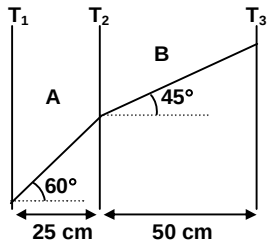
12. Which of the following options correctly represents the case when switch S_2 is closed and S_1 is open?
(A) (IV)(ii) (P) (B) (III) (i) (Q)
(C) (II) (i) (Q) (D) (II) (iii) (Q)

13. Which of the following options correctly represents the case when both the switches S_1 and S_2 is open?
(A) (I) (ii) (P) (B) (IV) (iv) (P)
(C) (I) (ii) (S) (D) (II) (iii) (Q)

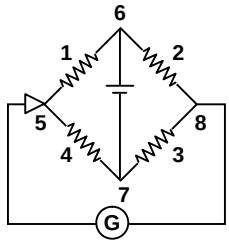
SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. Two conductors A and B each of cross section area 5 cm^2 are connected in series. Variation of temperature (in °C) along the length (in cm) is as shown in the figure. If thermal conductivity of A is $2\sqrt{3}$ J/m-sec-°C and the heat current in the conductors is 0.173 J/s. Find thermal conductivity of B (in J/m-sec-°C upto one decimal place.)



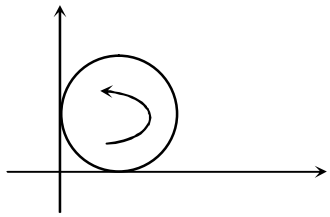
15. Four wires of equal length 8 m are arranged as shown in the figure. Wires, 2, 3 and 4 are of equal cross sectional area and wire 1 is of half the cross section of these wires. By how much distance pointer at point 5 must be moved to get null point?



Space for Rough work

16. Two particles of mass M = 3 kg each are kept on a horizontal circular platform on two mutually perpendicular radii at equal distance R = 1 m from the centre of the table. The particles are connected with a string, which is just taught when the platform is not rotating. Coefficient of friction between the platform and block is μ = 0.1. Then the maximum angular speed (ω in rad/sec) of platform about its centre so that the blocks remain stationary relative to platform. (take g = 10 m/s^2)

17. A uniform cylinder of radius R(= 3m) is spin about its axis at an angular velocity ω_0 (= $4\sqrt{10\pi}$ rad/sec) and placed between two perpendicular wall. The coefficient of friction between the walls and cylinder is μ(= 2). Then, how many turns will the cylinder make before it stops.



18. If voltage V = (100 $\pm$ 5) V and current I = (10 $\pm$ 0.2) A, the percentage error in resistance R is

Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **7** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Silicate having one monovalent corner oxygen atom in each tetrahedron unit is
(A) Sheet silicate (B) Cyclic silicate
(C) Single chain silicate (D) Double chain silicate

20. Choose incorrect statements:
(A) $AlCl_3$ undergoes hydrolysis to form $\left[Al(H_2O)_6\right]^{3+}$
(B) Amalgamated Al reacts with cold water to liberate H_2
(C) The ash formed on burning magnesium in air fumes on wetting and exposing to HCl
(D) B^{3+} ion can exist in aqueous solution in the hydrated form

21. Which of the following species exhibits the paramagnetic behaviour?
(A) NO (B) O_2^{2-}
(C) O_2^{2+} (D) O_2

22. Which of the following have identical bond order?
(A) CN^- (B) O_2
(C) NO^+ (D) CN^+

Space for Rough work

23. Given $\Delta H_f^0\left(C_2H_6, g\right) = -85\,kJ/mole$,

$\Delta H_f^0\left(C_3H_8, g\right) = -104\,kJ/mole$,

$\Delta H_{sub}^0\left(C, s\right) = 718\,kJ/mole$ & B.E.of $\left(H-H\right) = 436$ kJ/mole.

Then, in kJ/mole, the

(A) C – C bond enthalpy is 218 (B) C – H bond enthalpy is 414

(C) C – C bond enthalpy is 345 (D) C – H bond enthalpy is 448

24. Which of the following statement(s) is/are false?

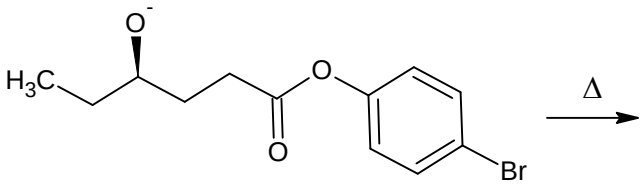
(A) All adiabatic processes are isoentropic (or isentropic) process

(B) The heat of vaporization of water at 100^0C is 40.6 kJ/mole. When 9g of water vapour condenses to liquid at 100^0C of 1 atm, then $\Delta S_{system} = 54.42\,J/K$

(C) When $\left(\Delta G_{system}\right)_{T,P} < 0$; the reaction must be exothermic

(D) dG = Vdp – S dT is applicable for closed system, both PV and non – PV work

25.



Which of the statements regarding the product formed in the above reaction is correct?

(A) If Br is replaced with NO_2 rate of reaction increases

(B) Inversion of configuration takes place.

(C) I Nucleophillic acyl substitution takes place.

(D) Configuration at chiral carbon does not change

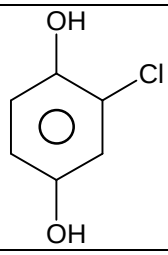
Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

Column-1 contains organic compounds treated with suitable reagent given in column-2 to give desired products listed in column-3. Now, answer the following questions

Column – 1		**Column – 2**		**Column – 3**	
(I)	Acetone	(i)	HCl	(P)	Pinacol
(II)	P-benzoquinone	(ii)	KOH(Fuse)	(Q)	Benzene+ $C_6H_5COO^-K^+$
(III)	Acetophenone	(iii)	(a) Mg/Benzene (b) H_2O	(R)	Benzoic acid
(IV)	Benzophenone	(iv)	(a) I_2 + NaOH (b) H^+/H_2O	(S)	

26. Which of the following is correct matching resulting into desired product as indicated?
(A) (IV), (i) & (S) (B) (III), (ii) & (R)
(C) (I), (iii) & (R) (D) (II), (i) & (S)

27. In which of the following sequence of reaction the diradical is formed as immediate.
(A) (II), (iv) & (R) (B) (III), (iv) & (R)
(C) (I), (iii) & (P) (D) (IV), (i) & (S)

28. Which of the following reaction sequence is correctly matched and does not involve redox reaction?
(A) (I), (iii) & (P) (B) (II), (i) & (S)
(C) (III), (iv) & (R) (D) (IV), (ii) & (Q)

Space for Rough work

Answer questions 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Column-1 contains compounds & column-2 & column-3. Now, answer the following questions

Column – 1		Column – 2		Column – 3	
(I)	Methyl isocyanate	(i)	sd^3-hybridisation	(P)	Reducing agent
(II)	Silyl isocyanate	(ii)	C-C bond length longer than 1.54 $\overset{o}{A}$	(Q)	Coloured due to charge transfer
(III)	Oxalic acid	(iii)	N is sp-hybridized	(R)	Linear
(IV)	$KMnO_4$	(iv)	N is sp^2 hybridized	(S)	V-shaped

29. Which of the following is correctly matched?
(A) (I), (iii) & (R) (B) (II), (iv) & (S)
(C) (III), (ii) & (P) (D) (IV), (i) & (S)

30. Which of the following statement is NOT correct for methyl isocyanate?
(A) (I), (iv) & (S) (B) (III), (i) & (P)
(C) (I), (ii) & (R) (D) (II), (iii) & (Q)

31. Which of the following is correctly matched?
(A) (II), (iii) & (R) (B) (II), (iv) & (R)
(C) (I), (iii) & (R) (D) (I), (i) & (S)

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. In alkaline medium, how many moles of potassium permanganate are required for complete oxidation of 3.5 moles of KI to KIO_3?

33. When H_2S is passed through a solution of $CoCl_2$, a black precipitate is formed. It is soluble in aqua regia. On adding acetic acid in excess and potassium nitrate, yellow precipitate is formed. The oxidation number of cobalt in this compound is +n. Find the value of n.

34. How many geometrical isomers are possible for the following complex?
$\left[Pt(NH_3)(NH_2OH)(NO_2)(py)\right]NO_2$

35. Dissociation constant for a weak base BOH is 10^{-5}. What is the value of pH when it is half neutralized by HCl?

36. The number types of monochloroalkanes formed by chlorination of sobutene is

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **7** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let P_n be the point on the complex plane with origin O expressed by $P_n = (1+i)\left(1+\frac{i}{\sqrt{2}}\right)\left(1+\frac{i}{\sqrt{3}}\right).....\left(1+\frac{i}{\sqrt{n}}\right)(n = 1,2,3,.....)$ then correct option(s) is (are)

(A) $\tan\left(\arg\left(\frac{P_{n+1}}{P_n}\right)\right) = \sqrt{\frac{n+2}{n+1}}$

(B) area of $\Delta OP_nP_{n+1} = \frac{\sqrt{n+1}}{2}$

(C) area of $\Delta OP_nP_{n+1} = \frac{\sqrt{n+3}}{2}$

(D) $\tan\left(\arg\left(\frac{P_{n+1}}{P_n}\right)\right) = \frac{1}{\sqrt{n+1}}$

38. Let $\vec{a},\vec{b},\vec{c}$ are vectors in xy plane such that $\vec{a}.\vec{c} = \vec{b}.\vec{c} = -\sqrt{3}\,\vec{a}.\vec{b}$ and $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$ then $\vec{a}.\vec{b}$ can be equal to

(A) $\frac{1}{2}$

(B) $\frac{-1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{-1}{3}$

Space for Rough work

39. Let f:[0, 1] $\to [0,\infty)$ be a real valued continuous function such that $(f(t))^2 \le 1 + 2\int_0^t f(u)\,du \forall\, t \in [0,1]$ then which of the following is (are) correct?

(A) $f\left(\frac{1}{2}\right) \le \frac{3}{2}$ (B) $f\left(\frac{3}{4}\right) \ge \frac{7}{4}$

(C) $\int_0^{3/4} f(u)\,du > \frac{33}{32}$ (D) $\int_0^{1/2} f(u) \le \frac{5}{8}$

40. Let $f:[0,1] \to R$ be a continuous and differentiable function such that for any $x, y \in [0,1]$, $xf(y) + yf(x) \le 1$, then which of the following option(s) is (are) correct

(A) $\int_0^1 f(x)\,dx \le \frac{\pi}{4}$ (B) $\int_0^1 f'(x)x^2 dx \ge f(1) - 1$

(C) $\int_0^1 xf'(x)\,dx \ge f(1) - \frac{\pi}{4}$ (D) $\int_0^1 f(x)\,dx \le \frac{\pi}{8}$

41. Let ABC be a triangle such that $\angle B = 2\angle C$. Side BC is extended by a segment CD equal to $\frac{BC}{3}$. Also $\angle BAD = \phi$. Then which of the following is (are) correct?

(A) $(AC)^2 - (AB)^2 = (AB)(BC)$ (B) $AD = \frac{1}{3}(3AB + 2BC)$

(C) $AD = \frac{1}{3}(3AB - 2BC)$ (D) Area of $\Delta ABC = \frac{1}{2}(BC)^2 \cot\left(\frac{\phi}{2}\right)$

Space for Rough work

42. Let N be the number of functions $f:\{1,2,3,.....,2018\}\to\{1,2,3,.....2018\}$ satisfying $f(j)<f(i)+j-i\ \forall$ integers I, j such that $1\le i<j\le 2018$. Then correct option(s) is/are

(A) $N\in\left(2\sum_{i=0}^{2017}\left({}^{2017}C_i\right)^2,\ 3\sum_{i=0}^{2017}\left({}^{2017}C_i\right)^2\right)$

(B) $N\in\left(\sum_{i=0}^{2017}\left({}^{2017}C_i\right)^2,\ 2\sum_{i=0}^{2017}\left({}^{2017}C_i\right)^2\right)$

(C) $N={}^{2018}C_{2018}+{}^{2019}C_{2018}+.....+{}^{4034}C_{2018}$

(D) $N={}^{2017}C_{2017}+{}^{2018}C_{2017}+.....+{}^{4034}C_{2017}$

43. Let $f:R\to R$ be a continuous function such that $f\left(f\left(f(x)\right)\right)=x\ \forall\ x\in R,$ then correct option(s) is (are)

(A) $y=f(x)$ is a bijective function

(B) $y=f(x)$ is strictly increasing function

(C) $y=f(x)$ is strictly decreasing function

(D) $y=f(x)$ intersects $y=x^2$ at exactly two distinct points.

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

$A(2,-1,4), B(1,0,-1), C(1,2,3), D(a,b,c)$ be the vertices of a parallelogram ABCD taken in order. Let π be the plane containing parallelogram ABCD. Let L be the line passing through P (2, 1, 0) and parallel to vector $\hat{i}+\hat{j}+\hat{k}$. Let F be the foot of perpendicular from P to Plane π and I is mirror image of P in plane π.

Column 1		Column 2		Column 3	
(I)	Coordinates of D : (2, 1, 8)	(i)	(i) Equation of line AB $\vec{r}=(1,0,-1)+\delta(1,-1,5)(\delta\in R)$	(P)	Distance between AB and CD is $2\sqrt{2}$
(II)	Coordinates of F : $\left(\frac{13}{27},\frac{19}{27},\frac{4}{27}\right)$	(ii)	(ii) Equation of line PF $\vec{r}=(-5,-1,1)+\delta(7,2,-1)(\delta\in R)$	(Q)	Acute angle between line L and plane π is $\cos^{-1}\left(\frac{8}{9\sqrt{2}}\right)$
(III)	Coordinates of I : $\left(\frac{-2}{27},\frac{11}{27},\frac{8}{27}\right)$	(iii)	(iii) Equation of projection of line L on plane π $\vec{r}=(1,0,-1)+\delta(1,-19,-31)(\delta\in R)$	(R)	Equation of plane, containing line L and perpendicular to plane π is $-3x+8y-5z-3=0$.
(IV)	Foot of Perpendicular from A to BC: $\left(2,\frac{9}{5},\frac{13}{5}\right)$	(iv)	(iv) Equation of mirror image of line L in plane π $\vec{r}=(1,0,-1)+\delta(29,-11,-35)$ $(\delta\in R)$	(S)	Equation of plane passing through P and parallel to plane π is $7x+2y-z-16=0$

44. Which of the following is correct combination?
(A) (I) (i) (Q) (B) (II) (iii) (S)
(C) (III) (i) (P) (D) (II) (ii) (P)

Space for Rough work

45. Which of the following is incorrect combination?
(A) (I) (i) (P) (B) (III) (ii) (S)
(C) (III) (iv) (S) (D) (II) (i) (Q)

46. Which of the following is correct combination?
(A) (I) (i) (R) (B) (I) (iv) (R)
(C) (IV) (iv) (S) (D) (III) (i) (S)

Answer questions 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

A tangent L is drawn at any point P in the first quadrant on the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$. It intersect the x - axis at A and y – axis at B, such that area of ΔOAB is minimum. A variable point $Q(\alpha,\beta)$ is taken on L from which pair of tangents QC and QD are drawn to the circle $x^2+y^2=12$ (O is origin)

Column 1		Column 2		Column 3	
(I)	Coordinate of A : $\left(4\sqrt{2},0\right)$	(i)	Minimum area of $\Delta OAB = 12$	(P)	Locus of circumcentre of ΔQCD is $3x+4y=6\sqrt{2}$
(II)	Coordinate of $P:\left(\frac{4}{\sqrt{2}},\frac{3}{\sqrt{2}}\right)$	(ii)	Number of points where L intersects circle $x^2+y^2=12$ is 2	(Q)	Equation of Line $L:3x+4y-12\sqrt{2}=0$
(III)	Coordinate of $B:\left(0,4\sqrt{2}\right)$	(iii)	Number of points where L intersects circle $x^2+y^2=12$ is 0	(R)	Equation of locus of curve from where perpendicular tangents can be drawn to given ellipse is $x^2+y^2=24$
(IV)	Point of concurrency of variable chord of contact CD of the circle $x^2+y^2=12$ is $\left(\frac{3}{\sqrt{2}},\frac{4}{\sqrt{2}}\right)$	(iv)	Number of points on the curve $x^2+y^2=25$, from where perpendicular tangents can be drawn to $x^2+y^2=12$ is infinite	(S)	Line $x-y-\sqrt{7}=0$ intersecting the ellipse after reflection will pass through $\left(-\sqrt{7},0\right)$

47. Which of the following is correct combination?
(A) (I) (i) (P) (B) (II) (iv) (S)
(C) (IV) (iii) (P) (D) (III) (iii) (S)

Space for Rough work

48. Which of the following is correct combination?
(A) (III) (i) (P) (B) (IV) (iv) (S)
(C) (I) (iii) (R) (D) (II) (ii) (Q)

49. Which of the following is incorrect combination?
(A) (IV) (i) (P) (B) (IV) (iv) (R)
(C) (II) (i) (P) (D) (I) (ii) (P)

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. Suppose that the circle $x^2 + y^2 - 4ax - 2ay + 4a^2 = 0 (a > 0)$ touches the lines $y = mx$ at A and B such that AB = 3. Then the value of $\frac{4a}{\sqrt{5}}$ is

51. Let (x, y, z) be an ordered triplet of real numbers, such that $x < 1, y < 2, z < 3$ and $x + \frac{y}{2} + \frac{z}{3} > 0$. Let (x_0, y_0, z_0) be the triplet (x, y, z) for which expression $(1-x)(2-y)(3-z)\left(x + \frac{y}{2} + \frac{z}{3}\right)$ is maximum, then the value of $\frac{z_0}{x_0 y_0}$ is

52. Number of ordered pair (x, y) of real numbers such that $\sin x + \sin y = \sin(x + y)$ and $|x| + |y| = 1$ is

Space for Rough work

53. Let 'ℓ' be a tangent line at the point $T\left(t,\frac{t^2}{2}\right)(t>0)$ on the curve $C: y=\frac{x^2}{2}$. A circle S touches the curve C and the x – axis. Denote by $(x(t),y(t))$ the center of circle S. If $\lim\limits_{r\to 0^+}\int\limits_r^1 x(t)\,y(t)\,dt=\frac{\tan\frac{3\pi}{8}}{6a}$, then the value of a is

54. Urn A contains four white balls and two red balls. Urn B contains three red balls and three black balls. An urn is randomly selected and then a ball inside of that urn is drawn. The process of selecting an urn and drawing out a ball is repeated, without replacing the first ball. Let 'p' be the probability that the first ball drawn was red, given that the second ball drawn was black, than 15p equals

Space for Rough work